



Glycerol Stock

Designing Life with AI

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Description

This method is used to preserve bacterial strains for long-term storage by mixing them with glycerol, a conservation agent.

Material

- Sterile glycerol (for preparing 20% and 50% glycerol solutions)
- Sterile liquid bacterial culture (if starting from liquid)
- Bacterial colonies on agar plates (if starting from colony)
- Sterile MilliQ or autoclaved water (for glycerol dilution)
- Cryogenic tubes

Steps

The following steps have to be under highly sterile conditions.

From a liquid culture

1. Prepare a 50% glycerol solution to use as a conservation agent.
2. In a labelled cryogenic tube, mix 800 µL of 50% glycerol solution with 800 µL of liquid bacterial culture.
3. Mix gently to homogenize.
4. Store at -70 °C for long-term preservation.

From a colony on a plate

1. Prepare a glycerol 20% solution.
2. Fill a cryogenic tube with 800 µl of the glycerol solution.
3. Pick a single colony with an inoculation loop from the plate.
4. Swirl it in the cryogenic tube.
5. Store it at -70 °C for long-term preservation.